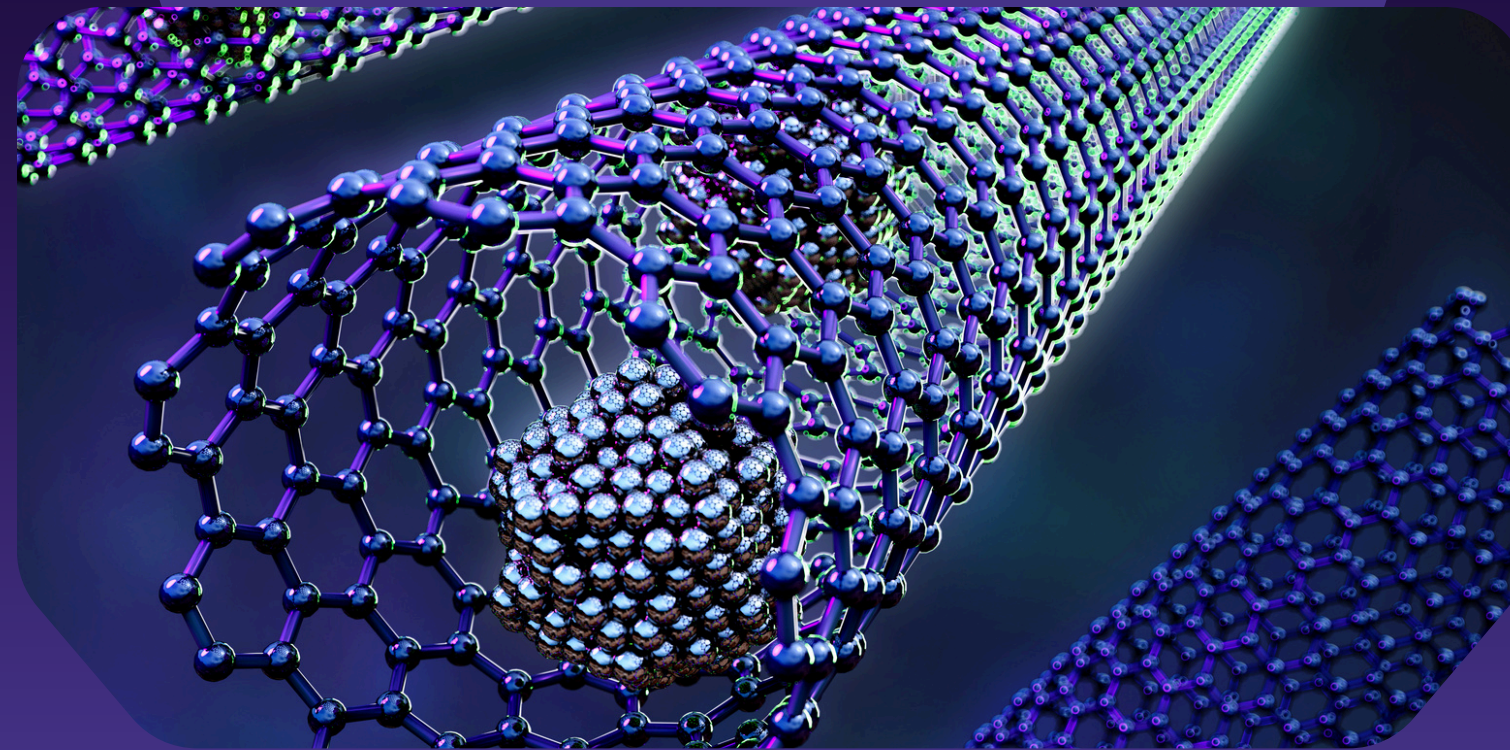




POLYMER AND NANOCOMPOSITES

INSTRUCTOR :- Prof. B K DHINDAW

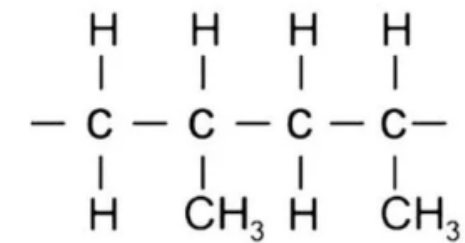
**Presented by :
HARISH PATIDAR
RAHUL M
K SATHVIKA**

POLYMERS

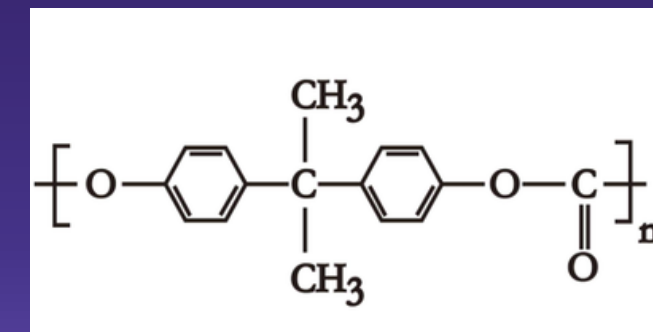


Trolley Bags

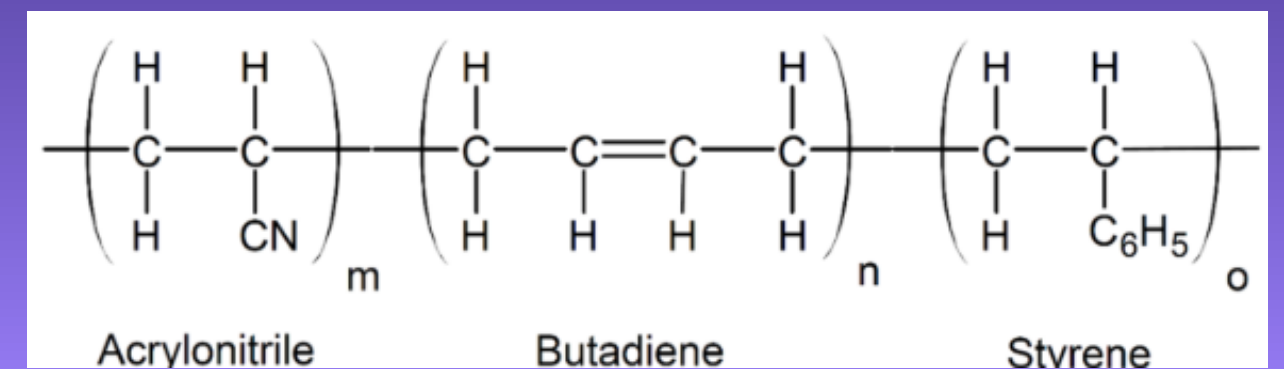
1) Polypropylene (PP)



2) Polycarbonate (PC)



3) Acrylonitrile Butadiene Styrene (ABS)



MONOMERS



1) Polypropylene (PP)

- Monomer (Propylene): $\text{CH}_2=\text{CH}-\text{CH}_3$

2) Polycarbonate (PC)

- Monomer 1 (Bisphenol A): $\text{HO}-\text{C}_6\text{H}_4-\text{C}(\text{CH}_3)_2-\text{C}_6\text{H}_4-\text{OH}$
- Monomer 2 (Phosgene): COCl_2

3) Acrylonitrile Butadiene Styrene (ABS)

- Monomer 1 (Acrylonitrile): $\text{CH}_2=\text{CH}-\text{C}\equiv\text{N}$
- Monomer 2 (1,3-Butadiene): $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
- Monomer 3 (Styrene): $\text{CH}_2=\text{CH}-\text{C}_6\text{H}_5$

Important properties

1) Polypropylene (PP)

- The lightest of all hard-shell luggage materials, offering an unmatched strength-to-weight ratio.
- Highly resistant to temperature fluctuations and chemicals.
- Ideal for frequent travelers who want durable protection while strictly avoiding overweight baggage fees.

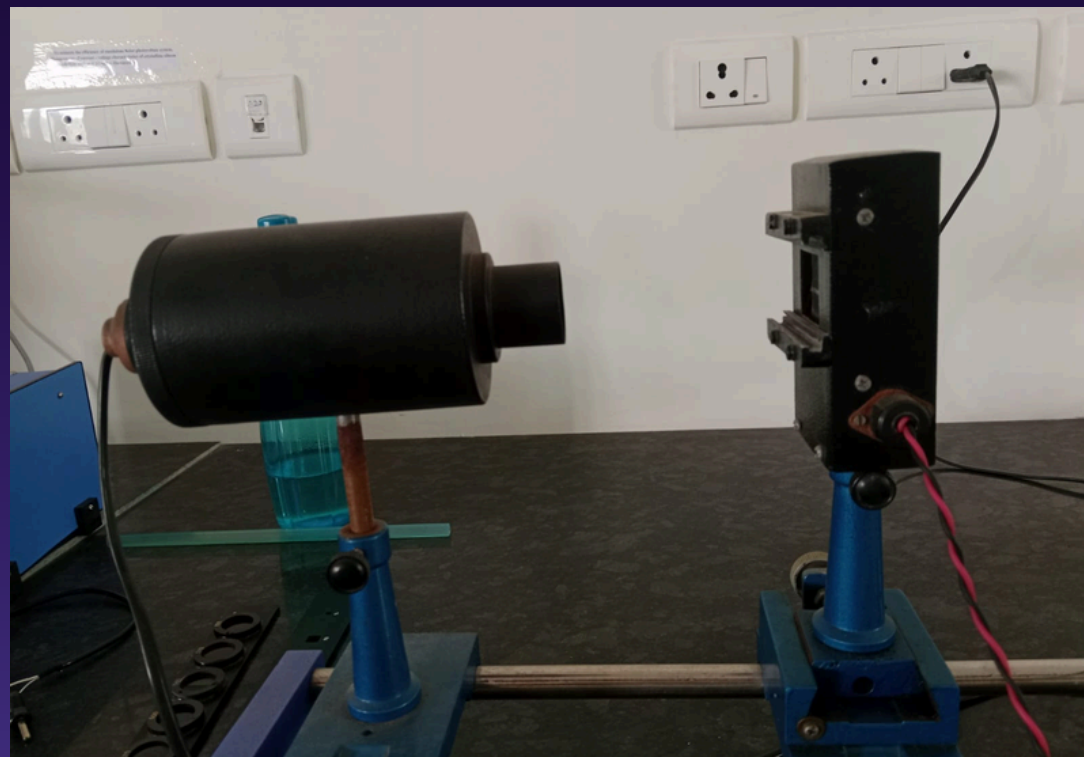
2) Polycarbonate (PC)

- Offers the highest impact resistance and overall durability.
- Highly flexible; the shell absorbs impact by bending or denting and easily pops back into its original shape without cracking.
- Best suited for premium luggage meant to withstand rough handling.

3) Acrylonitrile Butadiene Styrene (ABS)

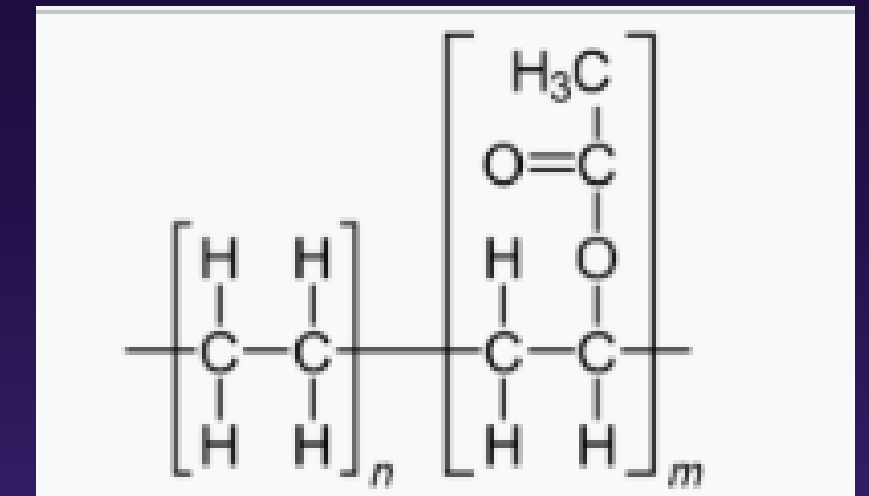
- Highly cost-effective to manufacture.
- Provides a rigid shell with excellent scratch resistance.
- Very easy to mold and dye, making it the standard for colorful, budget-friendly luggage designs.

SOLAR CELL PANELS



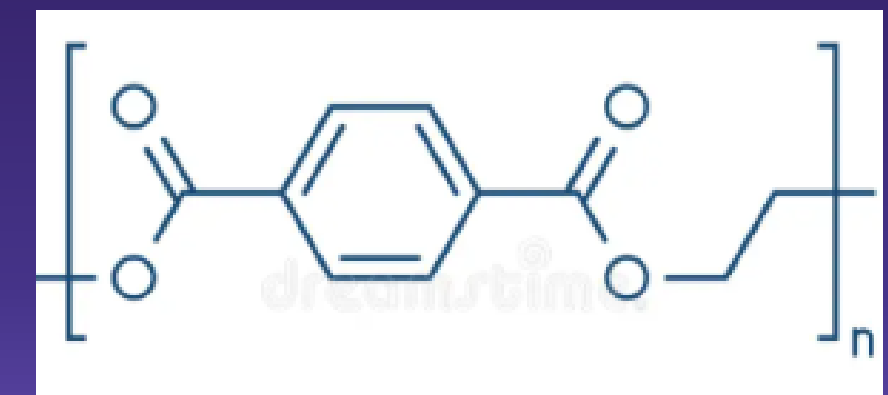
1. Ethylene Vinyl Acetate

- Ethylene: $\text{CH}_2=\text{CH}_2$
- Vinyl acetate: $\text{CH}_2=\text{CH}-\text{OCOCH}_3$



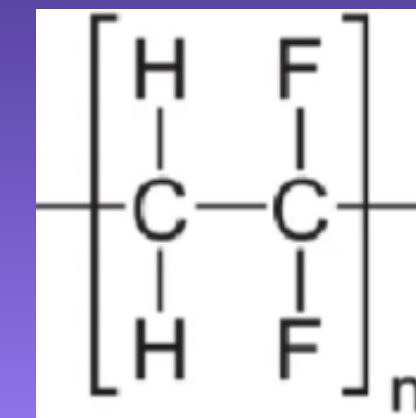
2. Polyethylene Terephthalate

- Ethylene glycol: $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$
- Terephthalic acid:
 $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$



3. Polyvinyl Fluoride

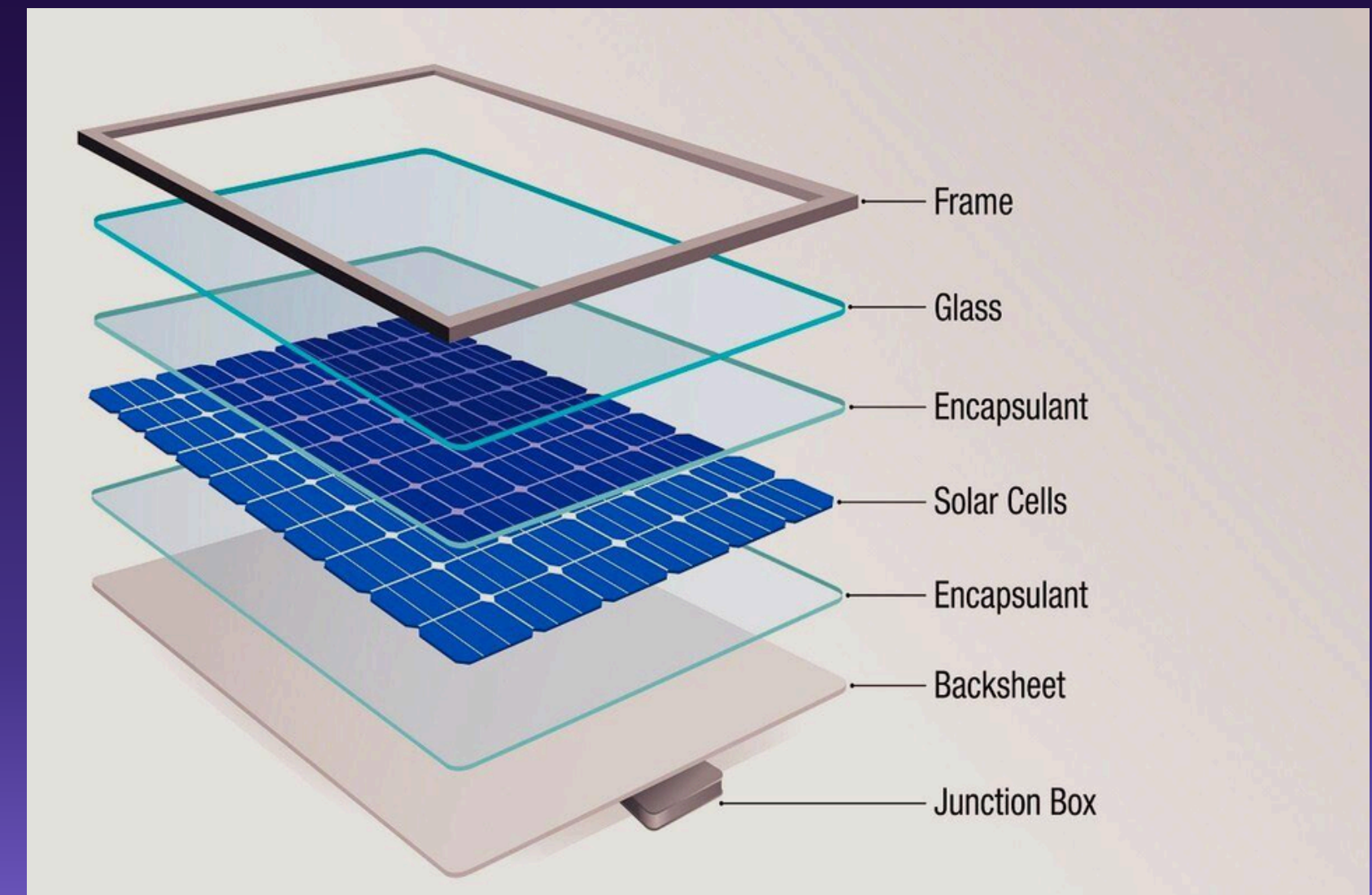
- vinyl Fluoride: $\text{CH}_2=\text{CHF}$



Important properties

1.EVA :Ethylene Vinyl Acetate

- **Transparent** → allows sunlight to reach the silicon cell
- **Flexible** → accommodates expansion/contraction due to temperature changes
- **Good adhesion** → bonds the cell to the surrounding layers
- **Moisture resistant** → helps protect the cell
- **Electrical insulation** → prevents unwanted electrical contact



Important properties

2.PET : Polyethylene Terephthalate

- This generally have High strength
- Lightweight and Flexible
- Good electrical insulation and Good dimensional stability
- PET is commonly used as a backsheet substrate, providing mechanical support and electrical insulation to the module.

3.PVF: Polyvinyl Fluoride

- Excellent UV resistance
- Excellent weather resistance and Chemical resistance
- Good durability and Electrical insulation

PVF is used as a protective outer layer of the backsheet, helping the solar module survive long-term exposure to sunlight, rain, humidity and outdoor conditions.

MEMORY FOAM MATTRESS

IIT BHUBANESWAR



Cover fabric

Polyester (PET) + elastane

Viscoelastic memory foam

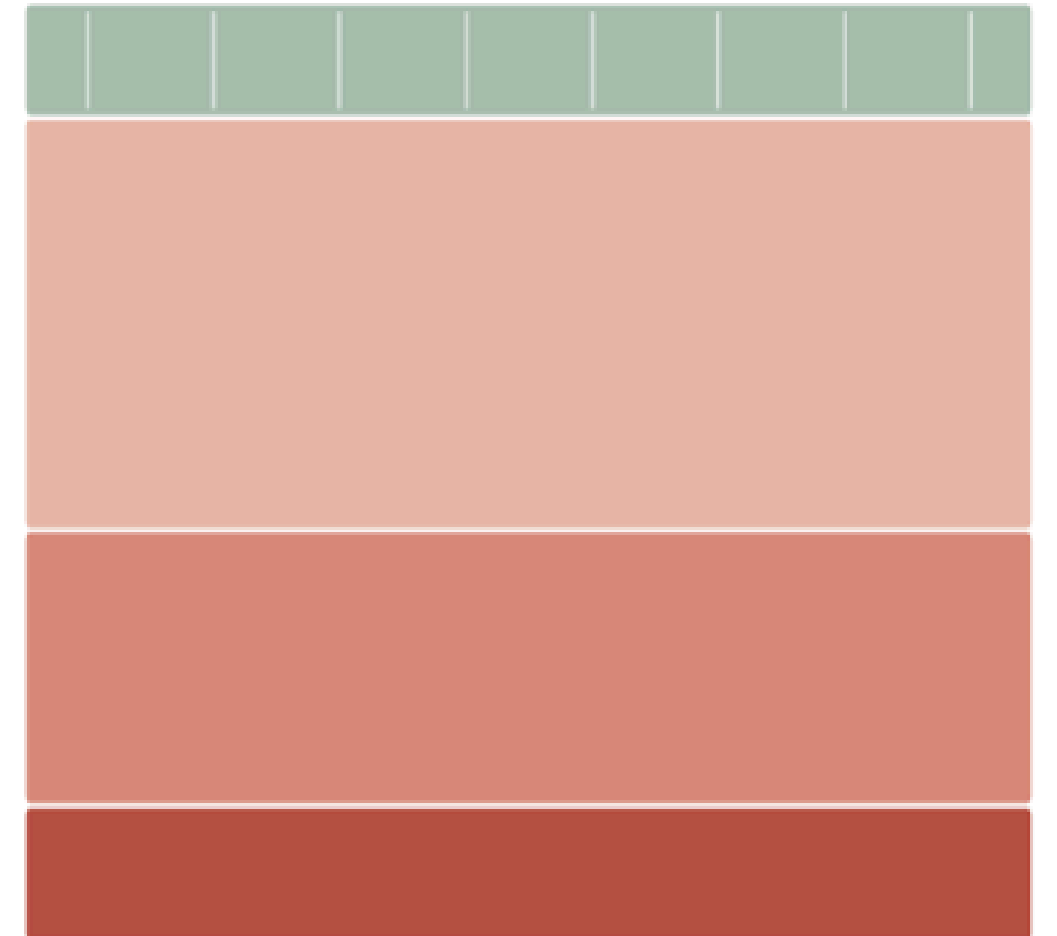
Polyether polyol + diisocyanate (TDI/MDI)

High-resilience support layer

Polyol + diisocyanate (TDI/MDI)

Base support foam

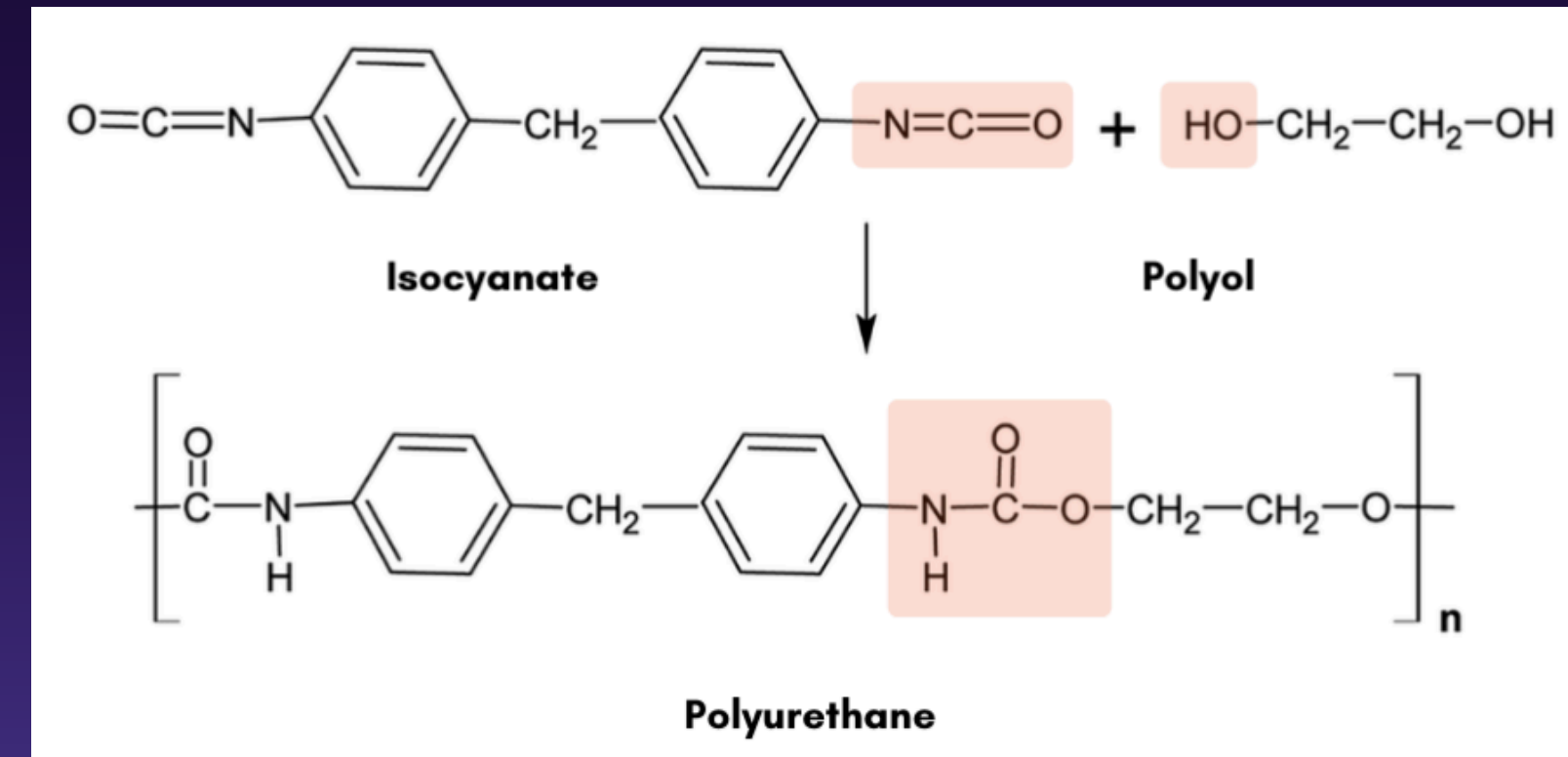
Conventional polyurethane



MONOMERS

1) Viscoelastic polyurethane(comfort layer)

- Polyether polyol
- Diisocyanate

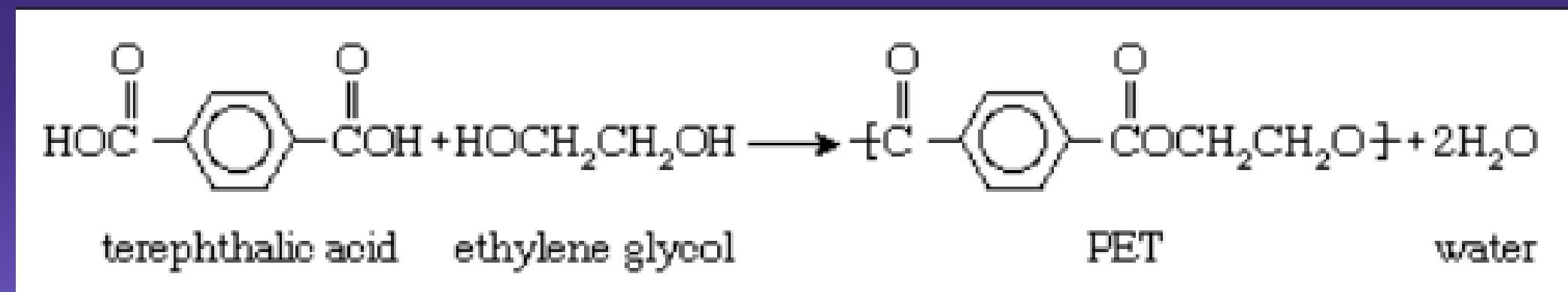


2) Conventional polyurethane(support base)

- Polyol
- Diisocyanate

3) Polyester / PET(cover fiber)

- Ethylene glycol
- Terephthalic acid



4) Elastane/spandex(cover stretch fiber)

- Macroglycol (polyol)
- Diisocyanate

Important properties

1) Viscoelastic Polyurethane(Memory Foam)

- Viscoelastic behavior → slowly deforms and recovers
 - Low resilience → reduces bounce
 - Excellent pressure distribution → conforms to the body
 - Open-cell structure → allows air movement
 - Temperature-sensitive response → improves conformity
- COMFORT + PRESSURE RELIEF

2) High-Resilience Polyurethane(Support Layer)

- High resilience → rapid recovery after compression
 - Good elasticity → responsive support
 - High fatigue resistance → withstands repeated compression
 - Good load-bearing capacity → supports upper layers
 - Low permanent deformation → maintains shape
- SUPPORT + DURABILITY

Important properties

3) Conventional Polyurethane(Base Support)

- **Lightweight** → reduces overall mattress weight
- **Good compressibility** → absorbs applied loads
- **Adequate stiffness** → provides structural support
- **Good fatigue resistance** → tolerates repeated loading
- **Cellular structure** → cushioning and insulation

→ **FOUNDATION + LOAD SUPPORT**

4) Polyester (PET) — Cover Fabric

- **High tensile strength** → resists tearing
- **Excellent abrasion resistance** → durable during use
- **Low moisture absorption** → dries relatively quickly
- **Good dimensional stability** → maintains fabric shape
- **Wrinkle resistance** → maintains appearance

→ **DURABLE + DIMENSIONALLY STABLE COVER**

Important properties

5) Elastane / Spandex — Cover Stretch Fiber

- Very high elastic recovery → returns close to original dimensions
 - High extensibility → accommodates large stretching
 - Excellent flexibility → conforms to mattress surface
 - Good fatigue resistance → withstands repeated stretching
 - Maintains a snug fit → prevents cover loosening
- **STRETCH + ELASTIC RECOVERY**

POLYURETHANE → comfort, cushioning & support | PET + ELASTANE → durable, flexible cover

THANK YOU